THE UNIVERSITY OF WESTERN AUSTRALIA

FINA2207

# BUSINESS ANALYSIS & VALUATION


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THE UNIVERSITY OF WESTERN AUSTRALIA

## Week 1

# Introduction to Business Analysis & Valuation

# Lecture 1 Outline

- Introduction to Financial statements
- Investing & Valuation
- Valuation: Project versus Enterprise

# INTRODUCTION TO FINANCIAL STATEMENTS

# The Four Financial Statements

1. Balance Sheet
2. Income Statement
3. Cash Flow Statement
4. Statement of Shareholders' Equity

# How Balance Sheet Components Fit Together

Assets = Liabilities + Shareholders' Equity

Or:

Shareholders' Equity = Assets – Liabilities

Compared to:

Value of Equity = Value of Firm – Value of Debt

# The Components of the Income Statement

Net Revenue – Cost of Goods Sold = Gross Margin

Gross Margin – Operating Expenses = **Operating Income (EBIT)**

Operating Income – Interest Expense + Interest Income = Income before taxes

Income before Taxes – Income Taxes = Income after Taxes and before Extraordinary Items

Income before Extraordinary Items + Extraordinary Items = Net Income

Net Income – Preferred Dividends = Net Income Available to Common shareholders

**Operating income** is sometimes called the **Earnings before Interest and Taxes** (EBIT)

# The Components of the Cash Flow Statement

Change in Cash = Cash from Operations

+ Cash from Investing

+ Cash from Financing

# The Components of the Equity Statement

***<u>The Stocks and Flow Equation:</u>***

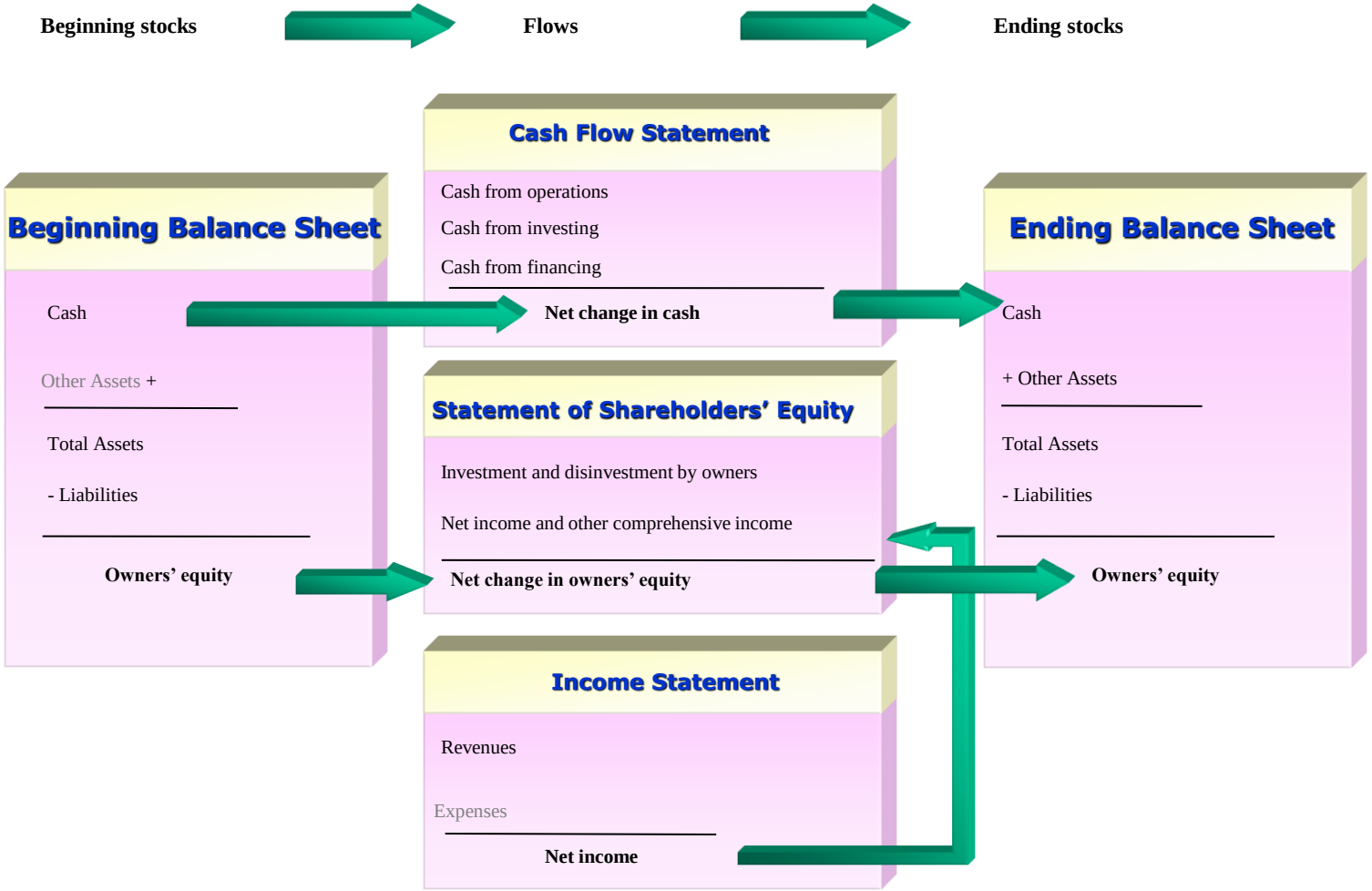
Ending equity =
Beginning equity + Total (comprehensive) income

– Net payout to shareholders

Comprehensive income =
Net income + Other comprehensive income

Net payout to shareholders =
Dividends + Share repurchases – Share issues

# The Articulation of the Financial Statements: How They Fit Together



# A Summary of the Accounting Relations

## How Parts of the Financial Statements Fit Together

**The Balance Sheet**

Assets
– Liabilities
= Shareholders' Equity

**The Income Statement**

Net Revenue
– Cost of Goods Sold
= Gross Margin
– Operating Expenses
= Operating Income before Taxes (EBIT)
– Net Interest Expense
= Income Before Taxes
– Income Taxes
= Income After Tax and before Extraordinary Items
+ Extraordinary Items
= Net Income
– Preferred Dividends
= Net Income Available to Common

**Cash Flow Statement (and the Articulation of the Balance Sheet and Cash Flow Statement)**

Cash Flow from Operations
+ Cash Flow from Investing
+ Cash Flow from Financing
= Change in Cash

**Statement of Shareholders' Equity (and the Articulation of the Balance Sheet and Income Statement)**

| | | |
|---|---|---|
| | | Dividends |
| | Net Income | + Share Repurchases |
| Beginning Equity | + Other Comprehensive Income | = Total Payout |
| + Comprehensive Income ← | = Comprehensive Income | – Share Issues |
| – Net Payout to Shareholders | ← | = Net Payout |
| = Ending Equity | | |

# Intrinsic Value and Book Value

- Intrinsic Premium:
  - Intrinsic Value of Equity – Book Value of Equity
- Market Premium:
  - Market Value of Equity – Book Value of Equity
- Intrinsic Price-to-Book Ratio:
  - $$\frac{\text{Intrinsic Value of Equity}}{\text{Book Value of Equity}}$$
- Price-to-Book Ratio:
  - $$\frac{\text{Market Value of Equity}}{\text{Book Value of Equity}}$$

# INVESTING & VALUATION

# Users of Firms' Financial Information (Demand Side)

- Equity Investors
  - *Investment analysis*
  - *Management performance evaluation*
- Debt Investors
  - *Probability of default*
  - *Determination of lending rates*
  - *Covenant violations*
- Management
  - *Strategic planning*
  - *Investment in operations*
  - *Evaluation of subordinates*
- Employees
  - *Security and remuneration*
- Litigants
  - ✓*Disputes over value in the firm*
- Customers
  - ✓*Security of supply*
- Governments
  - ✓*Policy making*
  - ✓*Regulation*
  - ✓*Taxation*
  - ✓*Government contracting*
- Competitors

**Investors and management are the primary users of financial statements**

# Investment Styles

- Intuitive Investing

  *-Rely on intuition and hunches: no analysis*

- Passive Investing

  *-Accept market price as value: no analysis*

  *-This is the "efficient market" approach*

- Fundamental Investing: Challenge market prices

  *-Active investing*

  *-Defensive investing*

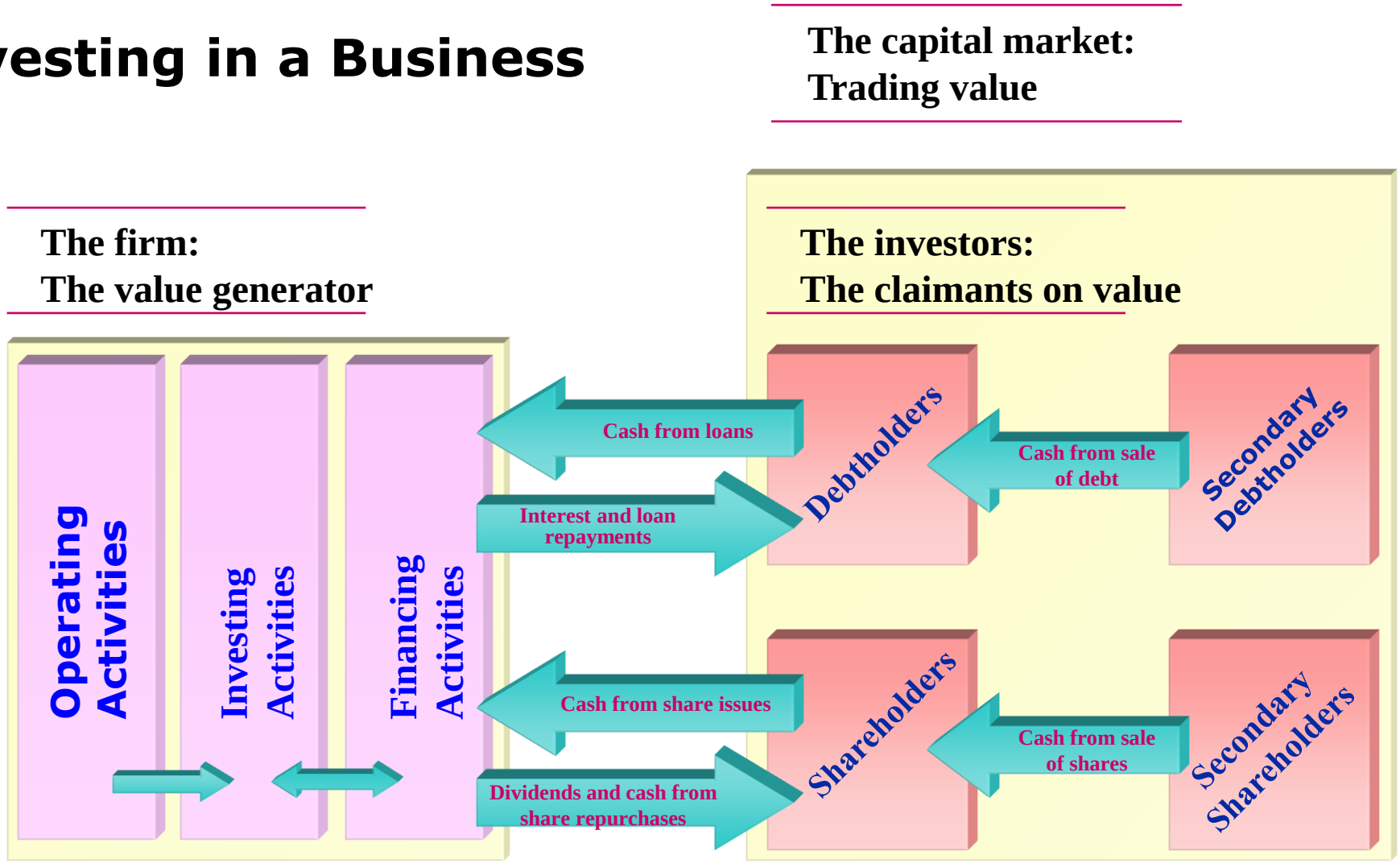
*A Motto for the Course*

*Price is what you pay, value is what you get*

# Fundamental Risk and Price Risk

- Fundamental risk is the risk that results from business operations
- Price risk is the risk of trading at the wrong price
  - *Paying too much*
  - *Selling for too little*

# Investing in a Business




Business investment and the firm: Value is surrendered by investors to the firm. The firm adds or loses value, and value is returned to investors. Financial statements inform about the investments. Investors trade in capital markets on the basis of information on financial statements.

# Business Activities

Financing Activities:

- Raising cash from investors and returning cash to investors

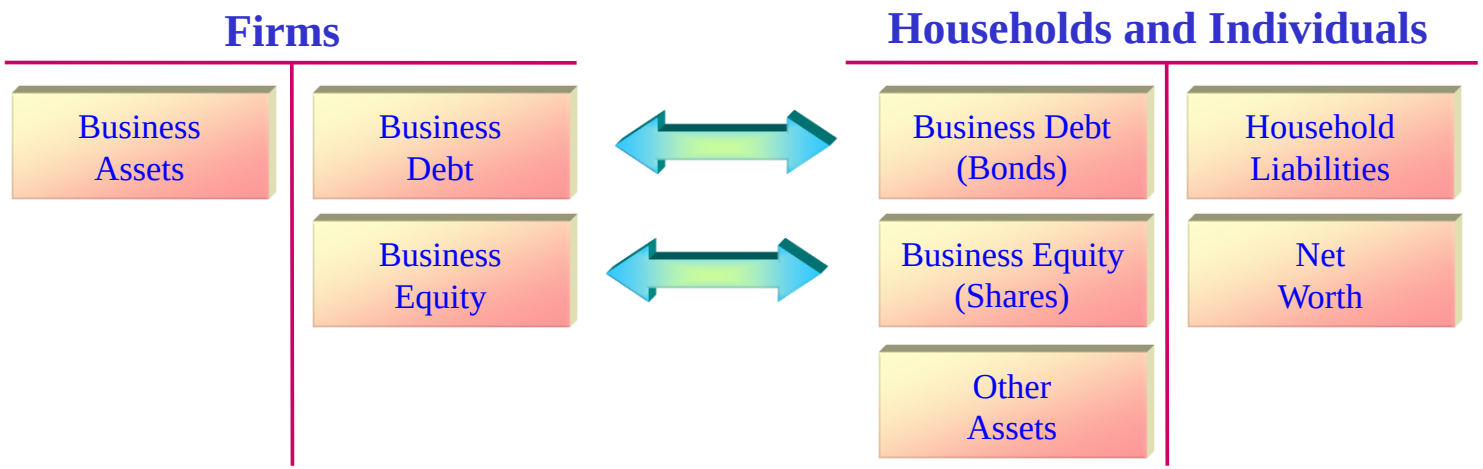
Investing Activities:

- Investing cash raised from investors in operational assets

Operating Activities:

- Utilizing investments to produce and sell products

# The Firm and Claims on the Firm




**<u>Value of the firm</u>** = Value of Assets
= Value of Debt +Value of Equity

$$V_0^F = V_0^D + V_0^E$$

Typically, valuation of debt is a relatively easy task.

# The Business of Analysis: The Professional Analyst

- The outside analyst understands the firm's value in order to advise outside investors
  - *Equity analyst*
  - *Credit analyst*

- The inside analyst evaluates plans to invest within the firm to generate value

- The <u>outside analyst</u> values the firm.
- The <u>inside analyst</u> values strategies for the firm

# The Analysis of Business

- **Understanding** the business is a necessary pre-requisite to carrying out a valuation
  - *Understand the business model (strategy)*
  - *Master the details*

- The financial statements are a lens on the business
- Financial statement analysis focuses the lens.

# Key Questions

- Does the firm have competitive advantage?
- How durable is the firm's competitive advantage?
- What forces are in play to promote competition?
- What protection does the firm have from competitors?

# Classifying and Ordering Information

*Don't Mix What You Know With Speculation*

- Order information in terms of how concrete it is: Separate concrete information from speculative information.
- Anchor a valuation on what you know rather than speculation.
- **Financial statements provide an anchor**.

# Anchoring Valuation in the Financial Statements

Value = **Anchor** + Extra Value

For example,

Value = Book value + Extra value

Value = Earnings + Extra Value

The valuation task:

**How to calculate the Extra Value**

# VALUATION:
# PROJECT versus ENTERPRISE

# Major Investment Decisions

*We will discuss how firms evaluate investments and grow and expand through:*

- **Project valuation**: firms acquire productive capacity by assembling necessary assets.
- **Enterprise valuation**: acquisitions of entire businesses - acquiring the productive assets of an existing firm
  - Common valuation tools and underlying principles can be used for both types of analysis.

# Project & Enterprise Valuation

*We will discuss how firms evaluate investments and grow and expand through:*

- Project Valuation
  - Rio Tinto joint venture agreement with CODELCO for copper exploration in Chile
  - Microsoft and HP 3-year $250 million investment into cloud computing
  - Coca-Cola's bottling and distribution joint-ventures in China
  - Fashion retailer Zara's (parent company Inditex) 77 market entries across the world including the most recent into India.
- Enterprise Valuation
  - Kraft Foods Inc. acquisition of Cadbury plc
  - Walt Disney Co's acquisitions of Marvel and Pixar
  - Coca-Cola Co. and PepsiCo's respective deals to acquire bottler operations
  - As of July 2010, Google has acquired 76 companies including Doubleclick and YouTube

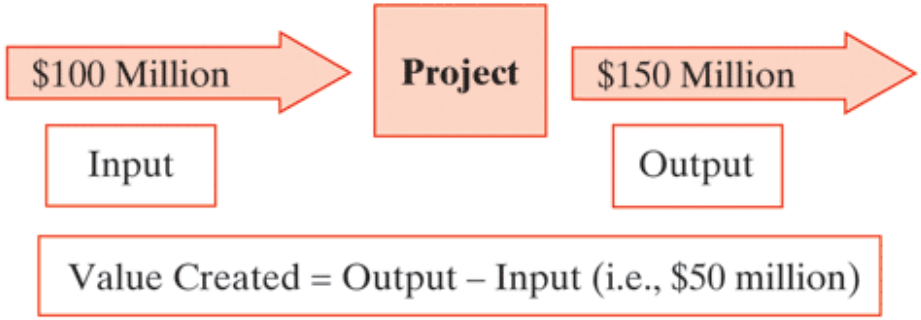
# Valuation: Tool for Value Creation

- The objective of a firm is to create wealth by initiating and managing investments that generate future cash flows that are worth more than the amount invested.
  - Management's goal is to avoid decision errors based on flawed or incomplete analysis
  - Valuation provides tools for the evaluation of new investment opportunities
    - From capital budgeting to mergers & acquisitions, valuation is more than discounting cash flows
    - Effective valuation analysis involves a disciplined 3-phase investment evaluation process

# Investments that Create Value

- Invest $100 million today in a project that generates a stream of cash flows valued at $150 million.
- Investment generates an incremental $50 million in wealth for its shareholders.
- The project has a net present value (NPV) of $50 million.[1]

**Figure 1-1** Investment Evaluation




[1]You will recall that NPV is equal to the difference between the value of expected future cash flows derived from an investment and the cost of making the investment.

# Investments that Destroy Value

- DaimlerChrysler formed by the $36 billion acquisition of Chrysler by Germany's Daimler-Benz in 1998
  - Net cash outflows, restructuring payments, Chrysler "bleeding cash and unlikely to become profitable under the best-case scenario until 2009"
- 2007 Cerberus Capital Management acquisition of 80% of Chrysler for $7.4 billion
  - According to one analyst, "Daimler is basically paying Cerberus to get rid of it."
  - More trouble in July 2007 with the debt capital market crunch
    - Trouble in syndication - J.P. Morgan Chase, Bear Stearns and Morgan Stanley asked Cerberus and DaimlerChrysler AG, to pay higher interest rates and change the convents.

Source: Cerberus 'rescues' Daimler" www.thedeal.com May-15-2007 and "Debt world trauma: Tricky talks" Jul-30-2007; by Lisa Gewirtz-Ward

# Investments that Destroy Value

- Over half of all large investment projects fail to achieve their hoped-for results.[1]
- Potential causes:
  - Managers "go with their gut"
  - Investments in risky projects
  - Uncertain future events
  - Incomplete information

[1]Nadim F. Matta and Ronald N. Ashkenas, 2003, Why good projects fail anyway, *Harvard Business Review* (September), 109–114.

# Project Valuation – Issues to Consider

- In any situation in which a company must value a major new investment, five key issues arise:

**Figure 1-3** **Important Issues to Think About When Making a Major Investment**

1. Does the "story" make sense?
2. What are the risks involved in undertaking the investment?
3. How can the investment be financed?
4. How does the investment affect near-term earnings?
5. Does the investment have inherent flexibilities?

# Enterprise Valuation

- Acquisitions involve similar issues, i.e Cisco's acquisition of Linksys.
  - Review the example provided in the chapter on Cisco/Linksys
- Practice Assignment: Research a recent acquisition and consider the framework of issues identified in the earlier examples.
  - Issue 1: Does the "story" make sense?
  - Issue 2: What are the sources of risk and can they be managed?
  - Issue 3: How can the investment be financed?
  - Issue 4: What is the short-term effect of the investment's acceptance on the firm's reported earnings?
  - Issue 5: Does the investment have inherent flexibilities that allow the firm to modify it in response to changing circumstances?
- Suggested acquisitions for consideration:
  - Oracle's acquisition of Sun Microsystems ($7.4 Billion)
  - Disney's acquisition of Marvel ($4 Billion)
  - Pfizer's acquisition of Wyeth ($68 Billion)

# Dealing with Complexity – Process & Discipline

*The three-phase investment evaluation process:*

**Figure 1-4** Three-Phase Investment Evaluation Process—Covering All the Bases

| | |
|---|---|
| **PHASE I** | **Investment (Idea) Origination and Analysis** |
| **Step 1:** | Conduct a strategic assessment. |
| **Step 2:** | Estimate the investment's value ("crunch the numbers"). |
| **Step 3:** | Prepare an investment evaluation report and recommendation to management. |
| **PHASE II** | **Managerial Review and Recommendation** |
| **Step 4:** | Evaluate the investment's strategic assumptions. |
| **Step 5:** | Review and evaluate the methods and assumptions used to estimate the investment's NPV. |
| **Step 6:** | Adjust for inherent estimation errors induced by bias, and formulate a recommendation regarding the proposed investment. |
| **PHASE III** | **Managerial Decision and Approval** |
| **Step 7:** | Make a decision. |
| **Step 8:** | Seek final managerial and possibly board approval. |

# Summing Up

- Process can be:
  - Very costly and time-consuming
  - Subject to biased estimates of project value such as conflicts of interest and incentive problems
  - Affected by problems arising out of differences in the information available to project champions and the internal review or control group (the strategic planning committee)

# Looking Forward

- Finance academics sometimes strip away complexities to focus on what determines value; sometimes creating a disconnect between what should be done in theory and what is done in practice.
- Our study of valuation integrates the analysis of individual projects and entire enterprises along two dimensions. Both build upon the same theoretical base.

# THANK YOU